

Axiom 10: Why is $1x = x$ for our closed scalar mult?

$$1x = \textcircled{x} = x. \quad \text{for all } x \in \mathbb{R}_+.$$

Axiom 8: For $c, d \in \mathbb{R}$, $v \in \mathbb{R}_+$
Compute $(c+d)v := v^{(c+d)} = \underline{v^c v^d}$

$$\underline{cv \oplus dv} := v^c \cdot v^d$$

Check axioms 7 and 9.

Subspaces and examples

If V is a vector space and W is a non-empty subset of V then we call W a subspace of V if W is closed under $+$ and scalar multiplication i.e. if $u, v \in W$ then $u + v \in W$ and if $c \in \mathbb{R}$ and $v \in W$ then $cv \in W$.

Example: Given an $m \times n$ matrix A , the set of solutions of $Ax = 0$ is a subspace of $\mathbb{R}^n$ called the *nullspace* of A , written $\text{Nul}(A)$.

$$\text{Nul}(A) = \{ x \in \mathbb{R}^n : Ax = 0 \} \text{ is a subspace of } \mathbb{R}^n.$$

How can we concretely see the $\text{Nul}(A)$?

Ex: Find the nullspace of $A =$

$$\rightarrow \begin{pmatrix} 2 & 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

We want all
 $x \in \mathbb{R}^6$ s.t.

$$Ax = 0.$$

$\text{rref}(A)$

$$\begin{pmatrix} 1 & 0 & 0 & \frac{3}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{4} & \frac{5}{4} \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

x_4, x_5, x_6 are free.

$$x_1 = -\frac{3}{2}x_4 + \frac{1}{2}x_5 - \frac{3}{2}x_6$$

$$x_2 = \frac{3}{4}x_4 - \frac{1}{4}x_5 - \frac{5}{4}x_6$$

$$x_3 = -x_5 - x_6$$

The solution
is then

$$x = \begin{pmatrix} -\frac{3}{2}x_4 + \frac{1}{2}x_5 - \frac{3}{2}x_6 \\ \frac{3}{4}x_4 - \frac{1}{4}x_5 - \frac{5}{4}x_6 \\ -x_5 - x_6 \\ x_4 \\ x_5 \\ \textcircled{x_6} \end{pmatrix}$$

for any
 x_4, x_5, x_6
 $\in \mathbb{R}$.

So

$\vec{x} =$
 $\uparrow$
 all $\vec{x}$
 in $\text{Nul}(A)$

$$x_4 \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

+

$$x_5 \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

+

$$x_6 \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

these vectors
 span $\text{Nul}(A)$

These vectors are also lin. indep.

The span as a subspace

Suppose that $v_1, \dots, v_k$ are vectors in $\mathbb{R}^n$. The span of $v_1, \dots, v_k$ is a subspace of $\mathbb{R}^n$. Recall that the span of $v_1, \dots, v_k$ is the set of all linear combinations of $v_1, \dots, v_k$. If $v_1, \dots, v_k$ are the columns of a matrix A then we call the span of these vectors the columnspace of A , written $\text{Col}(A)$.

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_k \\ | & & | \end{pmatrix} \quad \text{Col}(A) = \text{the span of } v_1, \dots, v_k.$$

How do we practically find a smallest spanning set for $\text{Col}(A)$?

Algorithm for Finding a spanning set for $\text{Col}(A)$ is to compute $\text{rref}(A)$ and pick the columns in A that correspond $\equiv$ to pivots in $\text{rref}(A)$. This set of columns is also lin. indep.

Examples of subspaces in general vector spaces

① $P_0 \subseteq P_1 \subseteq P_2 \subseteq P_3 \subseteq \dots \subseteq P \subseteq \text{cont functions.}$
 $\uparrow \quad \quad \quad \nearrow$
 subspaces.

② $W =$ all continuous functions s.t.

$$f(0) = 0.$$

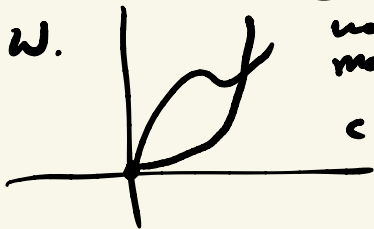
① zero function is in W .

② Closure under + :

$$0 + 0 = 0.$$

③ Closure under mult.

$c_0 = 0!$



Linear transformations between vector spaces

Definition

Suppose that V and W are vector spaces and $T: V \rightarrow W$ is a function from V to W . T is called a linear transformation from V to W if for all $u, v \in V$ and $c \in \mathbb{R}$,

$$T(u + v) = T(u) + T(v) \text{ and } T(cu) = cT(u).$$

The kernel and the range

Suppose that $T : V \rightarrow W$ is a linear transformation. Then we define

$$\ker(T) = \{v \in V : T(v) = 0\} \text{ and } \text{rng}(T) = \{T(v) : v \in V\}.$$